RELIANCE INDUSTRIES LIMITED
Jamnagar Refinery — Inspection & Corrosion Cell
EQUIPMENT INSPECTION REPORT
==
Inspection Report No: IR-NDT-JMR-2024-0291
Date of Inspection: 22-February-2024
Next Inspection Due: 22-August-2024 (6-monthly)
Inspection Type: Internal + Non-Destructive Testing (NDT)
----

1. EQUIPMENT UNDER INSPECTION

Equipment Tag: V-203
Equipment Type: Vertical Suction Scrubber (Knock-out Drum)
Service: Crude gas / liquid separation
Unit: VGO Hydrotreater Unit (VGO-HDT)
Design Code: ASME Section VIII, Division 1
Year of Commissioning: 2008
Last Inspection: 18-August-2023

----
2. DESIGN & OPERATING PARAMETERS

Design pressure: 22 kg/cm2g
Operating pressure: 16.5 kg/cm2g (measured during inspection: 16.2 kg/cm2g)
Design temperature: 180 degC
Operating temperature: 148 degC (measured: 151 degC)
Material of construction: SA-516 Gr. 70 (shell), SA-240 TP-316L (internals)
Nominal thickness (shell): 28 mm
Internal diameter: 1800 mm
Tangent-to-tangent length: 4800 mm

----
3. INSPECTION PERSONNEL & WITNESSES

Lead Inspector: K. Rajan (Senior Inspection Engineer, Level II — UT/MT/PT)
NDT Technician: Imran Khan (ASNT Level II — UT, MT, PT)
Reviewed by: G. Subramanian (Head, Inspection & Corrosion Cell)
Plant witness: Naresh Gupta (Unit Engineer, VGO-HDT)

----
4. NDT SURVEY CARRIED OUT
----
a) Ultrasonic Testing (UT) — Thickness Survey:
   Grid: 500 mm x 500 mm, 144 locations measured on shell and both heads.
   Minimum measured thickness: 26.1 mm (shell, south quadrant, elevation +2.0 m).
   Corrosion rate (general): 0.20 mm/year.
   Remaining corrosion allowance: 4.1 mm (design CA = 6.0 mm).
b) Magnetic Particle Testing (MT) — Shell welds and nozzle welds:
   Result: NO relevant indications. Acceptable.
c) Dye Penetrant Testing (PT) — Internal attachment welds (demister support):
   Result: Two linear indications, 3 mm and 5 mm long, at demister pad weld.
   Acceptance: Per ASME Section VIII, Div. 1, Appendix 8 — REPAIR REQUIRED.
d) Visual Inspection (VT):
   - Internal surface: light pitting (max 1.5 mm depth) on lower head.
   - Demister pad: partial blockage, requires cleaning.

- Inlet diverter: erosion at impingement plate, 2.0 mm wear groove.

----
5. FINDINGS & RISK CLASSIFICATION

Finding F-1: Demister support weld cracks (PT indications).
  Risk: HIGH. Repair weld + re-PT before return to service.
Finding F-2: Shell thinning to 26.1 mm (corrosion allowance reducing).
  Risk: MEDIUM. Re-rate or apply corrosion inhibitor; monitor quarterly.
Finding F-3: Inlet diverter impingement erosion.
  Risk: MEDIUM. Replace impingement plate at next turnaround (TA-2026).
Finding F-4: Light pitting on lower head.
  Risk: LOW. Monitor; record in RBI database.

----
6. REGULATORY & CODE REFERENCES

- OISD-164: Inspection of Pressure Vessels and Storage Tanks in Petroleum
  Industry (frequency and NDT coverage).
- Petroleum Rules 2002 (Schedule 7 — pressure vessels).
- Static and Mobile Pressure Vessels (Unfired) Rules (SMPV-U), 1981 —
  statutory inspection by PESO-approved competent person.
- Factories Act 1948, Section 28A (dangerous operations, pressure plant).
- ASME Section V (Nondestructive Examination) and Section VIII Div. 1.
- API 510: Pressure Vessel Inspection Code.
- API 572: Inspection of Pressure Vessels.

----
7. RECOMMENDATIONS
----
1. Repair demister support weld (Finding F-1) per WPS-QA-115 and re-PT
   before box-up. Responsible: Naresh Gupta. Target: 25-Feb-2024.
2. Continue injection of corrosion inhibitor at V-203 inlet; review
   injection rate with process engineering.
3. Carry out UT survey at 3-month intervals at the thin location;
   feed readings into RBI (Risk Based Inspection) tool.
4. Schedule impingement plate replacement in TA-2026.

----
8. FITNESS-FOR-SERVICE CONCLUSION

V-203 is Fit for Continued Service CONDITIONAL on completion of F-1 repair
and verification by re-PT. Re-inspection: 6-monthly UT survey, full internal
inspection at next turnaround (TA-2026).

==
Report digitally signed by Inspection Cell on 24-Feb-2024.
Distribution: VGO-HDT Unit, Reliability, Safety, PESO file.
==